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Feed train assemblies with skewed feeder roller pairs for a sugarcane harvester

Abstract

A feed train assembly for a sugarcane harvester includes a first feed section comprising a plurality of first pairs of upper and lower feed rollers configured to receive a first mat of sugarcane, with such pairs of upper and lower feed rollers comprising a first upstream pair of upper and lower feed rollers. The feed train assembly also includes a second feed section comprising a plurality of second pairs of upper and lower feed rollers configured to receive a second mat of sugarcane, and a combined feed section configured to receive the first and second mats of sugarcane from the first and second feed sections, respectively. A rotational axis of the lower feed roller of the first upstream pair of upper and lower feed rollers is oriented non-parallel relative to a rotational axis of the upper feed roller of the first upstream pair of upper and lower feed rollers.

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Background/Summary**FIELD OF THE INVENTION**

(1) The present subject matter relates generally to agricultural harvesters, such as sugarcane harvesters, and, more particularly, to a feed train assembly for a sugarcane harvester that includes one or more skewed or offset feeder roller pairs.

BACKGROUND OF THE INVENTION

(2) In an ever-changing agricultural landscape, adaptability is important to allow both manufacturers of agricultural harvesters and the end-users of such harvesters to be able to accommodate varying market demands, as well as varying trends in planting arrangements and/or

the like. The need for such adaptability is particularly relevant in the cultivation and harvesting of sugarcane and other tall, stalky plants, where the industry is undergoing a rapid evolution in terms of both the development of new varieties of plants and the use of varying planting configurations, all with an eye towards increased productivity. In this regard, manufacturers of sugarcane harvesters have made substantial efforts to provide machines that accommodate the varying market demands, such as by designing harvesters capable of harvesting two or more crop rows as opposed to a single row (i.e., multi-row harvesting). For example, U.S. Pat. No. 10,194,589, entitled "Multi-Row Sugarcane Harvester Feed Section," discloses a feed section for a sugarcane harvester with first and second feed trains for conveying respective mats of sugarcane deriving from separate rows to a downstream feed train for subsequent delivery to a chopper section of the harvester. Each feed train includes pairs of parallel upper and lower feed rollers, with each upper roller having a rotational axis that is parallel to the rotational axis of the lower roller with which it is paired. Such pairs of parallel upper and lower feed rollers are spaced apart from one another along the length of each feed train.

(3) While the above-described feed section provides an arrangement for harvesting two rows of sugarcane, further refinements and improvements are needed to accommodate the ever-changing market demands. For instance, the matched pairs of parallel upper and lower feed rollers spaced apart along the entire length of each feed train requires a longer overall feed section to convey the mats of sugarcane from the field to the chopper section, thereby resulting in a longer overall machine length.

(4) Accordingly, what is needed in the industry is a new feed train assembly that can be used within a multi-row sugarcane harvester that addresses one or more of the issues of the prior art.

BRIEF DESCRIPTION OF THE INVENTION

(5) Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

(6) In one aspect, the present subject matter is directed to a feed train assembly for a sugarcane harvester configured to simultaneously harvest first and second rows of sugarcane. The feed train assembly includes a first feed section configured to receive a first mat of sugarcane deriving from the first row of sugarcane, the first feed section comprising a plurality of first pairs of upper and lower feed rollers configured to convey the first mat of sugarcane from a first inlet of the first feed section to a first outlet of the first feed section. The plurality of first pairs of upper and lower feed rollers comprises a first upstream pair of upper and lower feed rollers positioned adjacent to the first inlet of the first feed section. The feed train assembly also includes a second feed section configured to receive a second mat of sugarcane deriving from the second row of sugarcane, with the second feed section comprising a plurality of second pairs of upper and lower feed rollers configured to convey the second mat of sugarcane from a second inlet of the second feed section to a second outlet of the second feed section. Additionally, the feed train assembly comprises a combined feed section positioned downstream of the first and second feed sections and configured to receive the first and second mats of sugarcane from the first and second feed sections, respectively. Moreover, a rotational axis of the lower feed roller of the first upstream pair of upper and lower feed rollers is oriented non-parallel relative to a rotational axis of the upper feed roller of the first upstream pair of upper and lower feed rollers.

(7) In another aspect, the present subject matter is directed to a feed train assembly for a sugarcane harvester configured to simultaneously harvest first and second rows of sugarcane. The feed train assembly includes a first feed section configured to receive a first mat of sugarcane deriving from the first row of sugarcane, with the first feed section comprising a plurality of first pairs of upper and lower feed rollers configured to convey the first mat of sugarcane from a first inlet of the first feed section to a first outlet of the first feed section. The plurality of first pairs of upper and lower feed rollers comprises a first upstream pair of upper and lower feed rollers positioned adjacent to the first inlet of the first feed section. The feed train assembly also includes a second feed section

configured to receive a second mat of sugarcane deriving from the second row of sugarcane, with the second feed section comprising a plurality of second pairs of upper and lower feed rollers configured to convey the second mat of sugarcane from a second inlet of the second feed section to a second outlet of the second feed section. The plurality of second pairs of upper and lower feed rollers comprises a second upstream pair of upper and lower feed rollers positioned adjacent to the second inlet of the first feed section. In addition, the feed train assembly includes a combined feed section positioned downstream of the first and second feed sections and configured to receive the first and second mats of sugarcane from the first and second feed sections, respectively. Moreover, a rotational axis of the lower feed roller of the first upstream pair of upper and lower feed rollers is oriented non-parallel relative to a rotational axis of the upper feed roller of the first upstream pair of upper and lower feed rollers, and a rotational axis of the lower feed roller of the second upstream pair of upper and lower feed rollers is oriented non-parallel relative to a rotational axis of the upper feed roller of the second upstream pair of upper and lower feed rollers.

(8) In a further aspect, the present subject matter is directed to a sugarcane harvester configured to simultaneously harvest first and second rows of sugarcane. The harvester includes a first base cutter assembly configured to sever sugarcane stalks of the first row of sugarcane and a second base cutter assembly configured to sever sugarcane stalks of the second row of sugarcane, with the first base cutter assembly defining a first central base cutter axis. The harvester also includes a feed train assembly positioned downstream of the first and second base cutter assemblies. The feed train assembly includes a first feed section configured to receive a first mat of severed sugarcane stalks from the first base cutter assembly, with the first feed section comprising a plurality of first pairs of upper and lower feed rollers configured to convey the first mat of severed sugarcane stalks from a first inlet of the first feed section to a first outlet of the first feed section. The plurality of first pairs of upper and lower feed rollers comprises a first upstream pair of upper and lower feed rollers positioned immediately downstream of the first base cutter assembly. The feed train assembly also includes a second feed section configured to receive a second mat of severed sugarcane stalks from the second base cutter assembly, with the second feed section comprising a plurality of second pairs of upper and lower feed rollers configured to convey the second mat of sugarcane from a second inlet of the second feed section to a second outlet of the second feed section. In addition, the feed train assembly includes a combined feed section positioned downstream of the first and second feed sections and configured to receive the first and second mats of severed sugarcane stalks from the first and second feed sections, respectively. Moreover, a rotational axis of the lower feed roller of the first upstream pair of upper and lower feed rollers is oriented non-parallel relative to a rotational axis of the upper feed roller of the first upstream pair of upper and lower feed rollers.

(9) These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

(2) FIG. 1 illustrates a schematic view of one embodiment of a multi-row sugarcane harvester in accordance with aspects of the present subject matter;

(3) FIG. 2 illustrates a top-down, schematic view of one embodiment of a feed train assembly

suitable for use with the harvester shown in FIG. 1 in accordance with aspects of the present subject matter;

(4) FIG. 3 illustrates a top view of one embodiment of a feed roller assembly suitable for use within the feed train assembly shown in FIG. 2 and/or the harvester shown in FIG. 1 in accordance with aspects of the present subject matter;

(5) FIG. 4 illustrates a bottom view of the feed roller assembly shown in FIG. 3;

(6) FIG. 5 illustrates a top view of an alternative embodiment of the feed roller assembly shown in FIGS. 3 and 4 in accordance with aspects of the present subject matter; and

(7) FIG. 6 illustrates a bottom view of the feed roller assembly shown in FIG. 5.

DETAILED DESCRIPTION OF THE INVENTION

(8) Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

(9) In general, the present subject matter is directed to an improved feed train assembly with skewed roller pairs for use within a multi-row sugarcane harvester. In several embodiments, the feed train assembly includes first and second feed sections for conveying respective mats of severed sugarcane stalks (e.g., deriving from separate sugarcane rows) from an upstream base cutter assembly to a downstream combined feed section for subsequent delivery of the merged mat of stalks to a chopper assembly of the harvester. Each of the first and second feed sections includes a plurality of pairs of upper and lower feed rollers. However, in contrast to known prior art arrangements, the rotational axes of at least one of the pairs of upper and lower rollers are skewed or offset from each other.

(10) In several embodiments, the furthest upstream pair of upper and lower rollers of each of the first and second feed trains is configured as a skewed pair of rollers. Specifically, the rotational axis of the lower roller of each upstream pair of rollers is skewed or offset relative to the rotational axis of the upper roller of each upstream pair of rollers. For example, in one embodiment, the rotational axis of the lower roller of each upstream pair of rollers may be oriented substantially perpendicular to a central base cutter axis of the adjacent base cutter assembly positioned immediately upstream of such roller, while the rotational axis of the upper roller of each upstream pair of rollers may be oriented substantially perpendicular to a skewed or angled feed axis (e.g., relative to the central base cutter axis) of the respective feed section. In such an embodiment, the lower roller of each upstream pair of feed rollers may function as a “butt-lifter” or “butt-lift” roller to lift the sugarcane severed by the adjacent base cutter assembly, while the upper roller of each upstream pair of feed rollers may function as a “steering” or “turning” rollers to initiate the “steering” or “turning” process in which the general flow direction of the mat of stalks is redirected from being substantially parallel to the respective central base cutter axis to being substantially parallel to the respective feed axis. Such dual-functionality of the upstream pair of rollers allows for the overall length of the feed train assembly (and, thus, the overall length of the machine) to be reduced, as the “steering” or “turning” process is initiated at the furthest upstream pair of rollers as opposed to the next adjacent downstream pair of feed rollers. For example, in some implementations, the skewed pair of upstream feed rollers may allow for the elimination of one pair of feed rollers from each of the first and second feed trains as compared to prior art arrangements.

(11) Referring now to the drawings, FIG. 1 illustrates a schematic view of one embodiment of a multi-row sugarcane harvester 10 for harvesting first and second rows 12, 14 of sugarcane. It should be appreciated that, for purposes of illustration, the harvester 10 is shown in FIG. 1 as

primarily including various components for cutting/harvesting, processing, cleaning, and discharging sugarcane as the cane is being harvested from an agricultural field. However, it should be appreciated that the harvester **10** may also include various other components, including drive components, traction components, hydraulic system components, and operator-related components. For instance, as is generally understood, the harvester **10** may include a frame or chassis, a pair of front traction devices (e.g., wheels or tracks), a pair of rear traction devices (e.g., wheels or tracks), and an operator's cab supported by the chassis. The harvester **10** may also include a primary source of power (e.g., an engine mounted on the chassis) which powers one or both pairs of the traction devices via a transmission, as well as one or more hydraulic fluid pumps configured to generate pressurized hydraulic fluid for powering the various hydraulic components of the harvester **10** (e.g., a plurality of hydraulic motors). Additionally, it should be appreciated that the harvester **10** may also include any other suitable components to facilitate cutting/harvesting, processing, cleaning, and/or discharging sugarcane.

(12) As particularly shown in FIG. **1**, the harvester **10** generally extends in lengthwise direction **16** between a forward end **18** and an aft end **20** along a central fore-to-aft or longitudinal axis **22**, which is generally oriented substantially parallel to the rows **12**, **14** being harvested. As shown, the harvester **10** includes first and second crop intake sections **24**, **26** positioned at its forward end **18** for intaking and cutting the sugarcane of the first and second rows **12**, **14**, respectively. Each crop intake section **24**, **26** includes a crop divider assembly **28** for gathering the sugarcane stalks of the respective rows **12**, **14** being harvested. As shown in FIG. **1**, each crop divider assembly **28** includes a pair of spiral crop divider rollers (CDRs), including an inner CDR **30** positioned closest to the central longitudinal axis **22** of the harvester **10** and an outer CDR **32** positioned further away from the central longitudinal axis **22**. The lateral spacing between the inner and outer CDRs **30**, **32** of each crop divider assembly **28** is generally selected such that the respective row **12**, **14** of sugarcane to be harvested is gathered between such CDRs **30**, **32** without intake of any adjacent sugarcane rows into the corresponding crop intake section **24**, **26**.

(13) Additionally, as shown in FIG. **1**, each crop intake section **24**, **26** includes one or more knock-down rollers (KDRs) **34** positioned within a throat of the intake section **24**, **26** defined between the CDRs **30**, **32** and a base cutter assembly (BCA) (e.g., a first base cutter assembly **36** of the first intake section **24** and a second base cutter assembly **38** of the second intake section **26**) positioned downstream of the knock-down rollers **34**. As the knock-down roller(s) **34** is rotated, the sugarcane stalks being harvested are knocked down while the CDRs **30**, **32** gather the stalks from the field into the throat of each respective intake section **12**, **14**. Each base cutter assembly **36**, **38** then severs the base of the downwardly angled stalks. As will be described below, each base cutter assembly **36**, **38** generally includes a pair of cutting discs with blades positioned along the outer periphery of each disc for severing the sugarcane stalks. The cutting discs are generally configured to be rotated via a hydraulic motor (not shown) powered by the vehicle's hydraulic system.

(14) It should be appreciated that, although not shown, the harvester **10** may also include one or more topper assemblies positioned at the forward end **18** of the harvester **10**, such as a single or common topper assembly for both rows **12**, **14** being harvested or a separate topper assembly for each respective row **12**, **14** being harvested. Each topper assembly may include, for example, a gathering disk and a cutting disc, with the gathering disk being configured to gather the sugarcane stalks so that the cutting disc may be used to cut off the top of each stalk. In addition, although not shown, the harvester **10** may also include a fin roller positioned behind the knock-down roller(s) **34**. For instance, the fin roller may include plurality of intermittently mounted fins that assist in further forcing the sugarcane stalks downwardly after being knocked down by the knock-down roller(s) **34**.

(15) As shown in FIG. **1**, the harvester **10** also includes a feed train assembly **40** located downstream of the crop intake sections **24**, **26** for conveying the severed stalks of sugarcane from each crop intake section **24**, **26** to a further downstream chopper assembly **60**. The feed train

assembly **40** includes first and second feed sections **42, 44** and a combined feed section **46** downstream of the first and second feed sections **42, 44**, with each feed section **42, 44, 46** including pairs of feed rollers (not shown) for conveying the severed sugarcane stalks received from the crop intake sections **24, 26** rearwardly to the chopper assembly **60**. For example, in several embodiments, the first and second feed sections **42, 44** are configured to receive a flow or mat of severed sugarcane stalks from the first and second crop intake sections **24, 25**, respectively and convey such separate flows or mats rearwardly into the combined feed section **46** for merging into a single merged flow or mat of sugarcane stalks. Specifically, upon being gathered, knocked down, and cut, the severed stalks from the first row **12** of sugarcane flow into the first feed section **42** and are directed rearwardly through such feed section **42** for delivery to the combined feed section **46**. Similarly, upon being gathered, knocked down, and cut, the severed stalks from the second row **14** of sugarcane flow into the second feed section **44** and are directed rearwardly through such feed section **44** for delivery to the combined feed section **46**. The merged flow or mat of sugarcane stalks directed through the combined feed section **46** is then fed downstream to the chopper assembly **60**.

(16) As will be described below, in several embodiments, the feed sections **42, 44, 46** may be configured to form a “y-shaped” feed train assembly **40**. For instance, in one embodiment, the combined feed section **46** may generally be centered along the longitudinal axis **22** of the harvester **10** while the various components of each crop intake section **24, 26** may generally be centered along respective first and second crop intake axes **48, 50** extending substantially parallel to, but spaced apart laterally from the longitudinal axis **22** (e.g., by a lateral distance equal to one-half the spacing defined between adjacent rows **12, 14**). In such an embodiment, the first and second feed sections **42, 44** may be angled inwardly relative to the longitudinal axis **22** such that the severed stalks delivered into each feed section **42, 44** from its respective intake section **24, 25** can be fed both rearwardly and inwardly (relative to the longitudinal axis **22**) towards the combined feed section **36**. For instance, as shown in FIG. 1, each of the first and second feed sections **42, 44** defines an inlet (e.g., a first inlet **52** of the first feed section **42** and a second inlet **56** of the second feed section **44**) immediately downstream of the adjacent base cutter assembly **36, 38** and an outlet (e.g., a first outlet **54** of the first feed section **42** and a second outlet **58** of the second feed section **44**) flowing into the combined feed section **46**. As shown in the illustrated embodiment, the inlet **52, 54** of each feed section **24, 26** is generally centered along the crop intake axis **48, 50** of its respective crop intake section **24, 26**, with the feed section **42, 44** being angled inwardly as it extends towards the combined feed section **46** such that the outlet **54, 58** of each feed section **42, 44** is positioned closer to the longitudinal axis **22** of the harvester **10** than the outlet **54, 58** of each feed section **42, 44**.

(17) It should be appreciated that, as used herein, a first component or axis is considered to be oriented substantially parallel to a second component or axis if the first component or axis is oriented relative to the second component or axis at an angle falling within an angular range of zero degrees plus-or-minus (+/-) 5 degrees, wherein an angle of zero degrees corresponds to a fully parallel orientation. Similarly, as used herein, a first component or axis is considered to be oriented substantially perpendicular to a second component or axis if the first component or axis is oriented relative to the second component or axis at an angle falling within an angular range of 90 degrees plus-or-minus (+/-) 5 degrees, wherein an angle of 90 degrees corresponds to a fully perpendicular orientation.

(18) Referring still to FIG. 1, the chopper assembly **60** is located at the downstream end of the feed train assembly **40** (e.g., adjacent to the downstream end of the combined feed section **46**). In general, the chopper assembly **60** is used to cut or chop the severed sugarcane stalks into pieces or “billets”, which may be, for example, six (6) inches long. The billets may then be propelled towards an elevator assembly **62** of the harvester **10** for delivery to an external receiver or storage device. Additionally, pieces of debris (e.g., dust, dirt, leaves, etc.) separated from the sugarcane

billets may be expelled from the harvester **10** through a primary extractor **64**, which is located immediately behind the chopper assembly **60** and is oriented to direct the debris outwardly from the harvester **10**. As is generally understood, the primary extractor **64** may include, for example, an extractor hood and an extractor fan mounted within the hood for generating a suction force or vacuum sufficient to pick up the debris and force the debris through the hood. The separated or cleaned billets, heavier than the debris being expelled through the extractor **64**, may then fall downward to the elevator assembly **62**. Moreover, as shown in FIG. **1**, the harvester **10** may also include an optional a secondary extractor **66** positioned at or adjacent to the distal end of the elevator assembly **62**. For example, the secondary extractor **66** may be located above a discharged opening of the elevator assembly **62**. In such an embodiment, upon being further “cleaned” by the secondary extractor **66**, the billets may then fall downwardly through the elevator discharge opening into an external storage device (not shown), such as a sugarcane billet cart.

(19) Referring now to FIG. **2**, a top-down, schematic view of one embodiment of the feed train assembly **40** described above with reference to FIG. **1** is illustrated in accordance with aspects of the present subject matter, particularly illustrating the various feed sections **42**, **44**, **46** of the feed train assembly **40** as well as the first and second base cutter assemblies **36**, **38** positioned at the downstream ends of the respective crop intake sections **24**, **26**. As indicated above, the first feed section **42** of the feed train assembly **40** is generally configured to receive a first flow or mat of severed sugarcane stalks (indicated by arrows **68**) from the first crop intake section **24** while the second feed section **44** of the feed train assembly **40** is generally configured to receive a second flow or mat of severed sugarcane stalks (indicated by arrows **70**) from the second crop intake section **26**. The separate flows or mats **68**, **70** of severed sugarcane stalks are then combined or merged within the combined feed section **46** into a single flow or mat of stalks (indicated by arrow **72**) for subsequent delivery to the downstream chopper assembly **60** (FIG. **1**) of the harvester **10**.

(20) As shown in FIG. **2**, each feed section **42**, **44**, **46** of the feed train assembly **40** generally includes a plurality of pairs **80** of feed rollers, with each pair **80** of feed rollers including an upper feed roller **82** and a lower feed roller **84**. Specifically, the first feed section **42** includes a plurality of first pairs **80A** of feed rollers (e.g., an upstream pair **80AU** of feed rollers, an intermediate pair **80AI** of feed rollers, and a downstream pair **80AD** of feed rollers), with each of such pairs **80A** including an upper feed roller **82A** and a lower feed roller **84A**. Similar, the second feed section **44** includes a plurality of second pairs **80B** of feed rollers (e.g., an upstream pair **80BU** of feed rollers, an intermediate pair **80BI** of feed rollers, and a downstream pair **80BD** of feed rollers), with each of such pairs **80B** including an upper feed roller **82B** and a lower feed roller **84B**. Additionally, the combined feed section **46** includes a plurality of pairs **80C** of feed rollers (e.g., an upstream pair **80CU** of feed rollers, an intermediate pair **80CI** of feed rollers, and a downstream pair **80CD** of feed rollers), with each of such pairs **80C** including an upper feed roller **82C** and a lower feed roller **84C**. To facilitate conveyance of the severed stalks through the various feed sections **42**, **44**, **46** of the feed train assembly **40**, each of the feed rollers **82**, **84** may generally be configured to rotate about a rotational axis (each axis being identified in FIG. **2** as the reference character for the associated fee roller **82**, **84** with an apostrophe (') added thereafter). For instance, in one embodiment, the feed rollers **82**, **84** may be configured to be rotationally driven (e.g., via respective hydraulic motors (not shown)) about their respective rotational axes.

(21) It should be appreciated that, for purposes of illustration, the upper feed rollers **82** are drawn with solid lines while the lower feed rollers **84** are drawn with dashed lines. Additionally, the lower feed rollers **84** are shown as being slightly smaller than the upper feed rollers **82** to allow the lower feed rollers **84** to be visible within the top-down view of FIG. **2**. However, it should be appreciated that the lower feed rollers **84** may have any size relative to the size of the upper feed rollers **82**, including being the same size of the upper feed rollers **82** or being larger than the upper feed rollers **84**. It should also be appreciated that, although the various feed sections **42**, **44**, **46** of the illustrated feed train assembly **40** are each shown as including three pairs **80** of feed rollers, each feed section

42, 44 may generally include any suitable number of pairs **80** of feed rollers, including less than three pairs **80** of feed rollers (e.g., two pairs of feed rollers) and greater than three pairs **80** of feed rollers (e.g., four or more pairs of feed rollers).

(22) Additionally, as shown in FIG. 2, each base cutter assembly **36, 38** generally includes a pair of cutting discs **74** configured to be rotated via a hydraulic motor and associated gearbox (not shown). Each cutting disc **74** is equipped with blades **76** positioned along its outer periphery for severing the sugarcane stalks. As shown in FIG. 2, each base cutter assembly **36, 38** generally defines a central base cutter axis **78, 79** (e.g., positioned equidistant from the rotational axes of the cutting discs **74**). For instance, the first base cutter assembly **36** of the first intake section **24** defines a first central base cutter axis **78**, while the second base cutter assembly **38** of the second intake section **26** defines a second central base cutter axis **79**. In one embodiment, each central base cutter axis **78, 79** may be oriented substantially parallel to (including being coaxial or substantially coaxial with) the associated intake axis **48, 50** (FIG. 1) of the respective crop intake section **24, 26**.

(23) As indicated above, in several embodiments, the feed train assembly **40** may be configured to have a “Y-shaped” arrangement or configuration. Specifically, as shown in FIG. 2, each of the first and second feed sections **42, 44** is generally oriented at an inwardly extending angle **86** relative to a central feed axis **88** of the combined feed section **46**, with such axis **88** being generally configured to extend substantially parallel to (including being coaxial or substantially coaxial with) the longitudinal axis **22** of the harvester **10**. For instance, the first feed section **42** generally extends lengthwise along a first feed axis **90** from a first inlet **52** (e.g., defined vertically between the upper and lower feed rollers **82A, 84A** of the upstream pair **80AU** of feed rollers of the first feed section **42**) to a first outlet **54** (e.g., defined vertically between the upper and lower feed rollers **82A, 84A** of the downstream pair **80AD** of feed rollers of the first feed section **42**), with the first inlet **52** being positioned immediately downstream of the adjacent base cutter assembly **36** of the first intake section **24** and the first outlet **54** being positioned immediately upstream and/or terminating at the combined feed section **46**. Similarly, the second feed section **44** generally extends lengthwise along a second feed axis **92** from a second inlet **56** (e.g., defined vertically between the upper and lower feed rollers **82B, 84B** of the upstream pair **80BU** of feed rollers of the second feed section **44**) to a second outlet **58** (e.g., defined vertically between the upper and lower feed rollers **82B, 84B** of the downstream pair **80BD** of feed rollers of the second feed section **44**), with the second inlet **56** being positioned immediately downstream of the adjacent base cutter assembly **38** of the second intake section **26** and the second outlet **58** being positioned immediately upstream and/or terminating at the combined feed section **46**. As shown in FIG. 2, the feed axis **90, 92** of each of the first and second feed sections **42, 44** is oriented at an acute feed angle **86** relative to the central feed axis **88** such that severed stalks received at the inlet **52, 56** of each feed section **42, 44** are directly rearwardly and inwardly (relative to the central feed axis **88**) towards the center of the combined feed section **46**. Thus, during operation, the first flow or mat **68** of severed stalks is generally directed past the base cutter assembly **36** of the first intake section **24** along the first central base cutter axis **78** (e.g., substantially parallel to the central feed axis **88**) until the flow/mat **68** of stalks enters the inlet **52** of the first feed section **42**, at which point the flow/mat **68** of stalks is turned inwardly for conveyance along the first feed axis **90** (e.g., at the feed angle **86** relative to the central feed axis **88**) towards the combined feed section **46**. Similarly, the second flow or mat **70** of severed stalks is generally directed past the base cutter assembly **38** of the second intake section **26** along the second central base cutter axis **78** (e.g., substantially parallel to the central feed axis **88**) until the flow/mat **70** of stalks enters the inlet **56** of the second feed section **44**, at which point the flow/mat **70** of stalks is turned inwardly for conveyance along the second feed axis **92** (e.g., at the feed angle **86** relative to the central feed axis **88**) towards the combined feed section **46**.

(24) Referring still to FIG. 2, in accordance with aspects of the present subject matter, one or more of the pairs **80** of feed rollers may correspond to or be configured as a skewed or offset pair of rollers, while the remainder of the pairs **80** of feed rollers each correspond to or are configured as a

parallel pair of rollers. In several embodiments, the lower feed rollers **82** positioned furthest upstream within the feed train assembly **40** are configured to be oriented non-parallel relative to the remainder of the feed rollers **82**, **84** included within its associated feed section **42**, **44** and/or non-parallel relative to the feed axis **90**, **92** of its associated feed section **42**, **44**. For example, as shown in FIG. 2, the rotational axis **84A'** of the lower roller **84A** of the upstream pair **80AU** of feed rollers of the first feed section **42** is oriented non-parallel relative to the rotational axes **82A'**, **84'** of the remainder of the feed rollers **82A**, **84A** included within the first feed section **42**. Specifically, in the illustrated embodiment, the rotational axis **84A'** of the lower roller **84A'** of the upstream pair **80AU** of feed rollers of the first feed section **42** is oriented substantially perpendicular to the first central base cutter axis **78** (and also substantially perpendicular to the central feed axis **88** and/or the first intake axis **48** (FIG. 1) of the first intake section **24**) while the rotational axes **82A'**, **84A'** of the remainder of the feed rollers **82A**, **84A** of the first feed section **42** are oriented substantially perpendicular to the feed axis **90** of the first feed section **42**. As such, in contrast to the intermediate and downstream pairs **80AI**, **80AD** of feed rollers of the first feed section **42** in which the rotational axes **82A'**, **84A'** of each roller pair are oriented substantially parallel to each other (and substantially perpendicular to the first feed axis **90**), the rotational axes **82A'**, **84A'** of the upper and lower feed rollers **82A**, **84A** of the upstream pair **80AU** are oriented non-parallel to each other (i.e., by being offset from each other by an acute offset angle **94**).

(25) Similarly, as shown in FIG. 2, the rotational axis **84B'** of the lower roller **84B** of the upstream pair **80BU** of feed rollers of the second feed section **44** is oriented non-parallel relative to the rotational axes **82B'**, **84B'** of the remainder of the feed rollers **82B**, **84B** included within the second feed section **44**. Specifically, in the illustrated embodiment, the rotational axis **84B'** of the lower roller **84B** of the upstream pair **84BU** of feed rollers of the second feed section **44** is oriented substantially perpendicular to the second central base cutter axis **79** (and also substantially perpendicular to the central feed axis **88** and/or the second intake axis **50** (FIG. 1) of the second crop intake section **26**) while the rotational axes **82B'**, **84B'** of the remainder of the feed rollers **82B**, **84B** of the second feed section **44** are oriented substantially perpendicular to the feed axis **92** of the second feed section **44**. As such, in contrast to the intermediate and downstream pairs **80BI**, **80BD** of feed rollers of the second feed section **44** in which the rotational axes **82B'**, **84B'** of each roller pair are oriented substantially parallel to each other (and substantially perpendicular to the second feed axis **92**), the rotational axes **82B'**, **84B'** of the upper and lower feed rollers **82B**, **84B** of the upstream pair are oriented non-parallel to each other (i.e., by being offset from each other by an acute offset angle **94**).

(26) The offset or skewed arrangement of the upstream pair **80AU**, **80BU** of feed rollers for each of the first and second feed sections **42**, **44** generally allows for enhanced transfer of the severed stalks from the adjacent base cutter assembly **36**, **38** to the respective feed section **42**, **44**, while also allowing the general flow direction of the flow/mat **68**, **70** of stalks entering each feed section **42**, **44** to be immediately reoriented or redirected along the respective feed axis **90**, **92**. Specifically, by orienting the lower roller **84A**, **84B** of each upstream pair **80AU**, **80BU** of feed rollers substantially perpendicular to the central base cutter axis **78**, **79** of the respective base cutter assembly **36**, **38**, such lower feed rollers **84A**, **84B** may function as “butt-lifters” or “butt-lift” rollers to lift the sugarcane severed by the adjacent base cutter assembly **36**, **38**. At the same time, by orienting the upper roller **82A**, **82B** of each upstream pair **80AU**, **80BU** of feed rollers substantially perpendicular to the feed axis **90**, **92** of the respective feed section **42**, **44** (and, thus, at the offset angle **94** relative to the respective lower roller **84A**, **84B**), such upper feed rollers **82A**, **82B** may function to initiate the “steering” or “turning” process in which the general flow direction of the flow/mat **68**, **70** of stalks is reoriented or redirected from being substantially parallel to the respective central base cutter axis **78**, **79** to being substantially parallel to the respective feed axis **90**, **92**. Accordingly, the upstream pair **80AU**, **80BU** of feed rollers for each of the first and second feed sections **42**, **44** serves a dual-role or otherwise provides dual functionality (i.e., both butt-

lifting functionality and turning functionality), which generally allows the feed sections **42**, **44** to be shortened in length relative to prior art configurations.

(27) Referring still to FIG. 2, as shown in the illustrated embodiment, the rotational axes **82C'**, **84C'** of the feed rollers **82C**, **84C** of the combined feed section **46** are all configured to be oriented substantially perpendicular to the central feed axis **88**. Specifically, the rotational axes **82C'** **84C'** of the upper and lower rollers **82C**, **84C** of each roller pair **80C** of the combined feed section **46** are oriented substantially perpendicular to the central feed axis **88**. As such, the combined or merged flow/mat **72** of stalks resulting from the separate respective flows/mats **68**, **70** of stalks received from the first and second feed sections **42**, **44** may be conveyed through the combined feed section **46** along the central feed axis **72** for subsequent delivery to the chopper assembly **60** (FIG. 1).

(28) Referring now to FIGS. 3 and 4, respective top and bottom views of one embodiment of a feed roller assembly **100** suitable for use within the disclosed harvester **10** and/or the disclosed feed train assembly **40** are illustrated in accordance with aspects of the present subject matter, particularly illustrating the feed roller assembly **100** positioned relative to a pair of cutting discs **74** of an upstream base cutter assembly. Specifically, for purposes of description, the feed roller assembly **100** is shown as including the upper and lower rollers **82A**, **84A** of the upstream pair **80AU** of feed rollers of the first feed section **42** (i.e., the pair of feed rollers **82A**, **84A** positioned closest to the adjacent base cutter assembly **36**) as well as the upper roller **82A** of the next adjacent pair of feed rollers of the first feed section **42** (i.e., the upper roller **82A** of the intermediate pair **80AI** of feed rollers). However, it should be appreciated that a similar configuration may also be utilized for the corresponding feed rollers **82B**, **84B** of the second feed section **44** (i.e., the upper and lower rollers **82B**, **84B** of the upstream pair **80BU** of feed rollers of the second feed section **44** and the upper roller **84B** of the intermediate pair **80BI** of feed rollers of the second feed section **44**).

(29) As particularly shown in the bottom view of FIG. 4, the rotational axis **84A'** of the lower roller **84** of the upstream pair **80AU** of feed rollers is oriented substantially perpendicular to the central base cutter axis **78** of the adjacent base cutter assembly **36**. In contrast, the rotational axes **82A'** of the upper rollers **82** shown in FIGS. 3 and 4 are oriented substantially perpendicular to the feed axis **90** of the associated feed section **42**. As such, the rotational axis **84'** of the lower roller **84'** of the upstream pair **80AU** of feed rollers is offset from the rotational axis **82'** of the upper roller **82** of the upstream pair **80AU** of feed rollers (and the upper roller **82** of the adjacent pair of feed rollers) by an offset angle **94**. Thus, as described above, the lower roller **84A** of the upstream pair **80AU** of feed rollers functions as a butt-lifter or butt-lift roller for lifting the sugarcane stalks severed by the adjacent base cutter assembly **36** while the upper roller **82A** of the upstream pair **80AU** of feed rollers functions as “turning” or “steering” roller that serves to reorient or redirect the flow of sugarcane stalks towards a flow path extending substantially parallel to the feed axis **90** of the associated feed section **42**.

(30) Referring now to FIGS. 5 and 6, respective top and bottom views of a feed roller assembly **100*** providing an alternative embodiment of the feed roller assembly **100** described above with reference to FIGS. 3 and 4 are illustrated in accordance with aspects of the present subject matter, particularly illustrating the feed roller assembly **100*** positioned relative to a pair of cutting discs **74** of an upstream base cutter assembly. For purposes of description, the feed roller assembly **100*** is shown as including the upper and lower rollers **82A**, **84A** of the upstream pair **80AU** of feed rollers of the first feed section **42** (i.e., the pair of feed rollers **82A**, **84A** positioned closest to the adjacent base cutter assembly **36**) as well as the upper and lower rollers **82A**, **84A** of the next adjacent pair of feed rollers of the first feed section **42** (i.e., the upper and lower rollers **82A**, **84A** of the intermediate pair of feed rollers **80AI**). However, it should be appreciated that a similar configuration may also be utilized for the corresponding feed rollers **82B**, **84B** of the second feed section **44** (i.e., the upper and lower rollers **82B**, **84B** of the upstream pair **80BU** of feed rollers of the second feed section **44** and the upper and lower rollers **82B**, **84B** of the intermediate pair **80BI**

of feed rollers of the second feed section 44).

(31) As shown, the upper rollers 82A of the feed roller assembly 100* are generally configured the same as that described above (e.g., by having rotational axes 82A' that are oriented substantially perpendicular to the feed axis 90 of the associated feed section 42). However, the rotational axes 84A' of the lower roller 84A of the upstream pair 80AU of feed rollers and the lower roller 84A of the adjacent intermediate pair 80AI of feed rollers are both oriented substantially perpendicular to the central base cutter axis 78 of the adjacent base cutter assembly 36. Thus, unlike the embodiment described above in which the rotational axis 84A' of the lower roller 84A' of the intermediate pair 80AI of feed rollers is configured to be oriented substantially perpendicular to the feed axis 90 of the associated feed section 42 (and substantially parallel to the rotational axis 82A' of the upper roller 82A of the intermediate pair 80AI of feed rollers), such lower roller 84A is oriented non-perpendicular to the feed axis 90 and non-parallel to the associated upper roller 82A. Specifically, as shown in FIG. 6, both lower rollers 84A are offset or skewed relative to their respective upper roller 82A by an offset angle 94.

(32) This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A feed train assembly for a sugarcane harvester configured to simultaneously harvest first and second rows of sugarcane, the feed train assembly comprising: a first feed section configured to receive a first mat of sugarcane deriving from the first row of sugarcane, the first feed section comprising a plurality of first pairs of upper and lower feed rollers configured to convey the first mat of sugarcane from a first inlet of the first feed section to a first outlet of the first feed section, the plurality of first pairs of upper and lower feed rollers comprising a first upstream pair of upper and lower feed rollers positioned adjacent to the first inlet of the first feed section, wherein the first feed section is configured to convey the first mat of sugarcane from the first inlet to the first outlet along a first feed axis of the first feed section; a second feed section configured to receive a second mat of sugarcane deriving from the second row of sugarcane, the second feed section comprising a plurality of second pairs of upper and lower feed rollers configured to convey the second mat of sugarcane from a second inlet of the second feed section to a second outlet of the second feed section; and a combined feed section positioned downstream of the first and second feed sections and configured to receive the first and second mats of sugarcane from the first and second feed sections, respectively, wherein a rotational axis of the lower feed roller of the first upstream pair of upper and lower feed rollers is oriented non-parallel relative to a rotational axis of the upper feed roller of the first upstream pair of upper and lower feed rollers, and wherein the rotational axis of the upper feed roller of the first upstream pair of upper and lower feed rollers is oriented substantially perpendicular to the first feed axis of the first feed section.

2. The feed train assembly of claim 1, wherein: the first mat of sugarcane is directed into the first inlet of the first feed section along a crop intake axis, and the rotational axis of the lower feed roller of the first upstream pair of upper and lower feed rollers is oriented substantially perpendicular to the crop intake axis.

3. The feed train assembly of claim 1, wherein: the first feed section is configured to convey the first mat of sugarcane from the first inlet to the first outlet along a first feed axis of the first feed section, the first mat of sugarcane is directed into the first inlet of the first feed section along a crop

intake axis, the rotational axis of the upper feed roller of the first upstream pair of upper and lower feed rollers is oriented substantially perpendicular to the first feed axis of the first feed section, and the rotational axis of the lower feed roller of the first upstream pair of upper and lower feed rollers is oriented substantially perpendicular to the crop intake axis.

4. The feed train assembly of claim 1, wherein: the plurality of second pairs of upper and lower feed rollers comprises a second upstream pair of upper and lower feed rollers positioned adjacent to the second inlet of the first feed section, and a rotational axis of the lower feed roller of the second upstream pair of upper and lower feed rollers is oriented non-parallel relative to a rotational axis of the upper feed roller of the second upstream pair of upper and lower feed rollers.

5. The feed train assembly of claim 1, wherein the first and second feed sections are arranged relative to the combined feed section such that the feed train assembly has a “Y-shaped” configuration.

6. The feed train assembly of claim 1, wherein the lower feed roller of the first upstream pair of upper and lower feed rollers is oriented relative to the first mat of sugarcane being received at the first inlet of the first feed section such that the lower feed roller functions as a butt-lift roller and the upper feed roller of the first upstream pair of upper and lower feed rollers is oriented relative to the first mat of sugarcane being received at the first inlet of the first feed section such that the upper feed roller functions as a turning roller that redirects the first mat of sugarcane along a first feed axis extending between the first inlet and the first outlet of the first feed section.

7. The feed train assembly of claim 1, wherein: the plurality of first pairs of upper and lower feed rollers further comprises a first intermediate pair of upper and lower feed rollers positioned immediately downstream of the first upstream pair of upper and lower feed rollers, and a rotational axis of the lower feed roller of the first intermediate pair of upper and lower feed rollers is oriented non-parallel relative to a rotational axis of the upper feed roller of the first intermediate pair of upper and lower feed rollers.

8. A feed train assembly for a sugarcane harvester configured to simultaneously harvest first and second rows of sugarcane, the feed train assembly comprising: a first feed section configured to receive a first mat of sugarcane deriving from the first row of sugarcane, the first feed section comprising a plurality of first pairs of upper and lower feed rollers configured to convey the first mat of sugarcane from a first inlet of the first feed section to a first outlet of the first feed section, the plurality of first pairs of upper and lower feed rollers comprising a first upstream pair of upper and lower feed rollers positioned adjacent to the first inlet of the first feed section; a second feed section configured to receive a second mat of sugarcane deriving from the second row of sugarcane, the second feed section comprising a plurality of second pairs of upper and lower feed rollers configured to convey the second mat of sugarcane from a second inlet of the second feed section to a second outlet of the second feed section, the plurality of second pairs of upper and lower feed rollers comprising a second upstream pair of upper and lower feed rollers positioned adjacent to the second inlet of the first feed section; and a combined feed section positioned downstream of the first and second feed sections and configured to receive the first and second mats of sugarcane from the first and second feed sections, respectively, wherein: a rotational axis of the lower feed roller of the first upstream pair of upper and lower feed rollers is oriented non-parallel relative to a rotational axis of the upper feed roller of the first upstream pair of upper and lower feed rollers, and a rotational axis of the lower feed roller of the second upstream pair of upper and lower feed rollers is oriented non-parallel relative to a rotational axis of the upper feed roller of the second upstream pair of upper and lower feed rollers.

9. The feed train assembly of claim 8, wherein: the first feed section is configured to convey the first mat of sugarcane from the first inlet to the first outlet along a first feed axis of the first feed section and the second feed section is configured to convey the second mat of sugarcane from the second inlet to the second outlet along a second feed axis of the first feed section, and the rotational axis of the upper feed roller of the first upstream pair of upper and lower feed rollers is oriented

substantially perpendicular to the first feed axis of the first feed section and the rotational axis of the upper feed roller of the second upstream pair of upper and lower feed rollers is oriented substantially perpendicular to the second feed axis of the second feed section.

10. The feed train assembly of claim 8, wherein: the first mat of sugarcane is directed into the first inlet of the first feed section along a first crop intake axis and the second mat of sugarcane is directed into the second inlet of the second feed section along a second crop intake axis, and the rotational axis of the lower feed roller of the first upstream pair of upper and lower feed rollers is oriented substantially perpendicular to the first crop intake axis and the rotational axis of the lower feed roller of the second upstream pair of upper and lower feed rollers is oriented substantially perpendicular to the second crop intake axis.

11. A sugarcane harvester configured to simultaneously harvest first and second rows of sugarcane, the harvester comprising: a first base cutter assembly configured to sever sugarcane stalks of the first row of sugarcane, the first base cutter assembly defining a first central base cutter axis; a second base cutter assembly configured to sever sugarcane stalks of the second row of sugarcane; and a feed train assembly positioned downstream of the first and second base cutter assemblies, the feed train assembly comprising: a first feed section configured to receive a first mat of severed sugarcane stalks from the first base cutter assembly, the first feed section comprising a plurality of first pairs of upper and lower feed rollers configured to convey the first mat of severed sugarcane stalks from a first inlet of the first feed section to a first outlet of the first feed section, the plurality of first pairs of upper and lower feed rollers comprising a first upstream pair of upper and lower feed rollers positioned immediately downstream of the first base cutter assembly; a second feed section configured to receive a second mat of severed sugarcane stalks from the second base cutter assembly, the second feed section comprising a plurality of second pairs of upper and lower feed rollers configured to convey the second mat of sugarcane from a second inlet of the second feed section to a second outlet of the second feed section, the plurality of second pairs of upper and lower feed rollers comprising a second upstream pair of upper and lower feed rollers positioned immediately downstream of the second base cutter assembly; and a combined feed section positioned downstream of the first and second feed sections and configured to receive the first and second mats of severed sugarcane stalks from the first and second feed sections, respectively, wherein a rotational axis of the lower feed roller of the first upstream pair of upper and lower feed rollers is oriented non-parallel relative to a rotational axis of the upper feed roller of the first upstream pair of upper and lower feed rollers, and wherein a rotational axis of the lower feed roller of the second upstream pair of upper and lower feed rollers is oriented non-parallel relative to a rotational axis of the upper feed roller of the second upstream pair of upper and lower feed rollers.

12. The harvester of claim 11, wherein: the first feed section is configured to convey the first mat of sugarcane from the first inlet to the first outlet along a first feed axis of the first feed section, and the rotational axis of the upper feed roller of the first upstream pair of upper and lower feed rollers is oriented substantially perpendicular to the first feed axis of the first feed section.

13. The harvester of claim 11, wherein the rotational axis of the lower feed roller of the first upstream pair of upper and lower feed rollers is oriented substantially perpendicular to the first central base cutter axis.

14. The harvester of claim 11, wherein: the first feed section is configured to convey the first mat of sugarcane from the first inlet to the first outlet along a first feed axis of the first feed section, the rotational axis of the upper feed roller of the first upstream pair of upper and lower feed rollers is oriented substantially perpendicular to the first feed axis of the first feed section, and the rotational axis of the lower feed roller of the first upstream pair of upper and lower feed rollers is oriented substantially perpendicular to the first central base cutter axis.

15. The harvester of claim 11, wherein the first and second feed sections are arranged relative to the combined feed section such that the feed train assembly has a “Y-shaped” configuration.

16. The harvester of claim 11, wherein the lower feed roller of the first upstream pair of upper and

lower feed rollers is oriented relative to the first base cutter assembly such that the lower feed roller functions as a butt-lift roller and the upper feed roller of the first upstream pair of upper and lower feed rollers is oriented relative to the first mat of sugarcane being received from the first base cutter assembly such that the upper feed roller functions as a turning roller that redirects the first mat of sugarcane along a first feed axis extending between the first inlet and the first outlet of the first feed section.

17. The harvester of claim 11, wherein: the plurality of the first pairs of upper and lower feed rollers further comprises a first intermediate pair of upper and lower feed rollers positioned immediately downstream of the first upstream pair of upper and lower feed rollers, and a rotational axis of the lower feed roller of the first intermediate pair of upper and lower feed rollers is oriented non-parallel relative to a rotational axis of the upper feed roller of the first intermediate pair of upper and lower feed rollers.

18. The harvester of claim 11, further comprising a chopper assembly positioned downstream of the combined feed section, the chopper assembly being configured to receive a merged mat of severed sugarcane stalks comprising a merged flow of the first and second mats of severed sugarcane stalks.
